Unsupervised Learning Lab Assignment

Objective

To programmatically architect a high-dimensional synthetic dataset and apply a complete unsupervised pipeline—**Standardization, PCA, and K-Means**—to recover hidden structures without the aid of labels.

Task 1: Synthetic Data Generation

Generate 5,000 samples with 784 features representing 28 * 28 pixel images across 10 distinct classes.

1. **Base Centroids:** Initialize 10 unique base centroids using random distributions.
2. **Variations:** Create 500 variations per class by adding Gaussian noise to the base centroids.
3. **Data Range:** Flatten the images into 784-dimensional vectors and ensure values are clipped within the [0, 255] range.
4. **Standardization:** Apply `StandardScaler` to the dataset to ensure each feature has a mean of 0 and a variance of 1.

Task 2: Dimensionality Reduction (PCA)

1. **Variance Retention:** Apply PCA to the standardized data, retaining **95% of the variance**.
2. **Analysis:** Record the number of components required to reach this threshold.
3. **Visualization:** Project the data onto the first two principal components (PC_1 and PC_2) using a scatter plot, color-coding by the original (hidden) class labels.

Task 3: Unsupervised Clustering (K-Means)

1. **Clustering:** Apply K-Means with (K=10) on the **PCA-reduced data**.
2. **Label Mapping:** Match the resulting cluster assignments with the true labels to compute the **Cluster Purity**.
3. **Visualization:** Compare the True Label scatter plot against the K-Means Cluster plot side-by-side.

Task 4: Interpretation & Reconstruction

1. **Scree Plot:** Generate a plot of cumulative explained variance to justify the PCA cutoff.
2. **Centroid Recovery:** Reshape the 10 K-Means centroids back into 28 * 28 images.
3. **Visual Verification:** Compare these reconstructed centroids to your original "Base Centroids" from Task 1 to see how much signal was preserved.